

101. If $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} =$

$Ax^3 + Bx^2 + Cx + D$, then $B + C$ is equal to :

यदि $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} =$

$Ax^3 + Bx^2 + Cx + D$ है, तो $B + C$ is बराबर है :

(A) -3

(B) 1

(C) 9

(D) -1

Ans. A

Sol. $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1$$

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & -x+2 & -2x+3 \\ 1 & -x+1 & x-5 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$$

$$C_2 \rightarrow C_2 - 2C_1, C_3 \rightarrow C_3 - 3C_1$$

$$\begin{vmatrix} x-2 & 1 & 2 \\ 1 & -x & -2x \\ 1 & -x-1 & x-8 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$$

$$-3x^3 + 12x^2 - 15x + 6 = Ax^3 + Bx^2 + Cx + D$$

$$B = 12, C = -15$$

$$B + C = -3$$

102. Let S be the set of all integer solutions, (x, y, z) of the system of equations

$$x - 2y + 5z = 0$$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

such that $15 \leq x^2 + y^2 + z^2 \leq 150$. Then, the number of elements in the set S is equal to _____.

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$$x - 2y + 5z = 0$$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

के पूर्णांकीय हलों (x, y, z) का समुच्चय S है, जिनके लिए $15 \leq x^2 + y^2 + z^2 \leq 150$, तो S के अवयवों की संख्या है _____।

Ans. 8

Sol. $x - 2y + 5z = 0$

$$-2x + 4y + z = 0$$

$$-7x + 14y + 9z = 0$$

$$D = \begin{vmatrix} 1 & -2 & 5 \\ -2 & 4 & 1 \\ -7 & 14 & 9 \end{vmatrix}$$

$$D = 1(36 - 14) + 2(-18 + 7) + 5(-28 + 28)$$

$$D = 22 - 22 + 0 = 0$$

Let $y = k$

$$x - 2k + 5z = 0$$

$$-2x + 4k + z = 0$$

$$z = 0$$

$$x = 2k$$

solutions

(x, y, z)

$(2k, k, 0)$

$$s \leq x^2 + y^2 + z^2 \leq 150$$

$$s \leq 4k^2 + k^2 + 0^2 \leq 150$$

$$1 \leq k^2 \leq 30$$

$$K = \pm 5, \pm 4, \pm 3, \pm 2$$

hence total number of solutions 8.

103. If the minimum and the maximum values of the function $f: \left[\frac{\pi}{4}, \frac{\pi}{2} \right] \rightarrow \mathbb{R}$, defined by

$$f(\theta) = \begin{vmatrix} -\sin^2 \theta & -1 - \sin^2 \theta & 1 \\ -\cos^2 \theta & -1 - \cos^2 \theta & 1 \\ 12 & 10 & -2 \end{vmatrix} \text{ are } m \text{ and } M \text{ respectively, then the ordered pair } (m, M) \text{ is equal to :}$$

$$\text{यदि } f(\theta) = \begin{vmatrix} -\sin^2 \theta & -1 - \sin^2 \theta & 1 \\ -\cos^2 \theta & -1 - \cos^2 \theta & 1 \\ 12 & 10 & -2 \end{vmatrix} \text{ द्वारा परिभाषित फलन } f: \left[\frac{\pi}{4}, \frac{\pi}{2} \right] \rightarrow \mathbb{R} \text{ के निम्नतम तथा उच्चतम मान क्रमशः } m$$

तथा M हैं, तो क्रमित युग्म (m, M) बराबर हैं :

- (A) $(0, 2\sqrt{2})$ (B) $(0, 4)$ (C) $(-4, 4)$ (D) $(-4, 0)$

Ans. D

Sol. $C_2 \rightarrow C_2 - C_1 + C_3$

$$f(\theta) = \begin{vmatrix} -\sin^2 \theta & 0 & 1 \\ -\cos^2 \theta & 0 & 1 \\ 12 & -4 & -2 \end{vmatrix} = 4 \cos 2\theta$$

$$\theta \in \left[\frac{\pi}{4}, \frac{\pi}{2} \right]$$

$$\Rightarrow 4 \cos 2\theta \in [-4, 0]$$

104. If $a + x = b + y = c + z + 1$, where a, b, c, x, y, z are non-zero distinct real numbers, then $\begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix}$ is

equal to :

$$\text{यदि } a + x = b + y = c + z + 1 \text{ हैं, जहाँ } a, b, c, x, y, z \text{ शून्येतर भिन्न वास्तविक संख्याएँ हैं, तो } \begin{vmatrix} x & a+y & x+a \\ y & b+y & y+b \\ z & c+y & z+c \end{vmatrix} \text{ बराबर है}$$

:

- (A) 0 (B) $y(b-a)$ (C) $y(a-b)$ (D) $y(a-c)$

Ans. C

Sol. $C_3 \rightarrow C_3 - C_1$

$$= \begin{vmatrix} x & a+y & a \\ y & b+y & b \\ z & c+y & c \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_3$$

$$= \begin{vmatrix} x & x & a \\ y & y & b \\ z & y & c \end{vmatrix}$$

$$= y \begin{vmatrix} x & 1 & a \\ y & 1 & b \\ z & 1 & c \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2, \text{ then } R_2 \rightarrow R_2 - R_3$$

$$= y \begin{vmatrix} x-y & 0 & a-b \\ y-z & 0 & b-c \\ z & 1 & c \end{vmatrix}$$

$$= y(-1)[(x-y)(b-c) - (y-z)(a-b)]$$

$$= -y((b-c)(b-a) - (a-b)(c-b-1))$$

$$= y(a-b)$$

105. If the system of linear equations

$$x + y + 3z = 0$$

$$x + 3y + k^2z = 0$$

$$3x + y + 3z = 0$$

has a non-zero solution (x, y, z) for some $k \in \mathbb{R}$, then $x + \left(\frac{y}{z}\right)$ is equal to :

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$$x + y + 3z = 0$$

$$x + 3y + k^2z = 0$$

$$3x + y + 3z = 0$$

का, किसी $k \in \mathbb{R}$, के लिए, एक शून्येतर हल (x, y, z) है, तो $x + \left(\frac{y}{z}\right)$ बराबर है :

(A) -9

(B) -3

(C) 3

(D) 9

Ans. B

Sol. For non-zero solution

$$D = 0$$

$$\begin{vmatrix} 1 & 1 & 3 \\ 1 & 3 & k^2 \\ 3 & 1 & 3 \end{vmatrix} = 0$$

$$1(9 - k^2) - (3 - 3k^2) + 3(1 - 9) = 0$$

$$9 - k^2 - 3 + 3k^2 - 24 = 0$$

$$2k^2 - 18 = 0$$

$$k^2 = 9$$

system of equation

$$x + y + 3z = 0 \dots\dots\dots(1)$$

$$x + 3y + 9z = 0 \dots\dots\dots(2)$$

$$3x + y + 3z = 0 \dots\dots\dots(3)$$

(A) $-(3)$

$$2x = 0, \Rightarrow x = 0$$

$$\frac{y}{z} = -3$$

106. Let m and M be respectively the minimum and maximum values of

$$\begin{vmatrix} \cos^2 x & 1 + \sin^2 x & \sin 2x \\ 1 + \cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1 + \sin 2x \end{vmatrix}$$

Then the ordered pair (m, M) is equal to :

माना m तथा M

$$\begin{vmatrix} \cos^2 x & 1 + \sin^2 x & \sin 2x \\ 1 + \cos^2 x & \sin^2 x & \sin 2x \\ \cos^2 x & \sin^2 x & 1 + \sin 2x \end{vmatrix} \text{ के, क्रमशः}$$

न्यूनतम तथा अधिकतम मान है, तो क्रमित युग्म (m, M) बराबर है :

- (A) (1, 3) (B) (-3, 3) (C) (-4, -1) (D) (-3, -1)

Ans. D

Sol. $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$D = \begin{vmatrix} \cos^2 x & 1 + \sin^2 x & \sin 2x \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{vmatrix}$$

$$D = -2 - \sin 2x$$

$$m = -3$$

$$M = -1$$

107. The sum of distinct values of λ for which the system of equations

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$$

has non-zero solutions, is _____.

λ के उन भिन्न मानों का योग, जिनके लिए समीकरण निकाय

$$(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$$2x + (3\lambda + 1)y + 3(\lambda - 1)z = 0$$

के शून्येतर (non-zero) हल हैं, है _____।

Ans. 3

Sol. $(\lambda - 1)x + (3\lambda + 1)y + 2\lambda z = 0$

$$(\lambda - 1)x + (4\lambda - 2)y + (\lambda + 3)z = 0$$

$$2x + (3\lambda + 1)y + (3\lambda - 3)z = 0$$

$$\begin{vmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ \lambda - 1 & 4\lambda - 2 & \lambda + 3 \\ 2 & 3\lambda + 1 & 3\lambda - 3 \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{vmatrix} 0 & -\lambda+3 & +\lambda-3 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$(\lambda-3) \begin{vmatrix} 0 & -1 & 1 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$\lambda = 3 \text{ or } \begin{vmatrix} 0 & -1 & 1 \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3\lambda-3 \end{vmatrix} = 0$$

$$C_2 \rightarrow C_2 + C_3$$

$$\begin{vmatrix} 0 & 0 & 1 \\ \lambda-1 & 5\lambda+1 & \lambda+3 \\ 2 & 6\lambda-2 & 3\lambda-3 \end{vmatrix} = 0$$

$$((\lambda-1)(6\lambda-2) - 2(5\lambda+1)) = 0$$

$$6\lambda^2 - 8\lambda + 2 - 10\lambda - 2 = 0$$

$$\lambda^2 - 3\lambda = 0$$

$$\lambda(\lambda-3) = 0$$

$$\Rightarrow \lambda = 0, 3 \quad \text{sum} = 0 + 3 = 3$$

108. For the system of linear equations :

$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in \mathbb{R}$$

Consider the following statements :

- (A) The system has unique solution if $k \neq 2, k \neq -2$.
- (B) The system has unique solution if $k = -2$.
- (C) The system has unique solution if $k = 2$.
- (D) The system has no-solution if $k = 2$.
- (E) The system has infinite number of solutions if $k \neq -2$.

Which of the following statements are correct ?

(A) (C) and (D) only

(B) (A) and (E) only

(C) (B) and (E) only

(D) (A) and (D) only

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$$x - 2y = 1, x - y + kz = -2, ky + 4z = 6, k \in \mathbb{R}$$

के लिए, नीचे दिए कथनों पर विचार कीजिए :

(A) निकाय का केवल एक हल है, यदि $k \neq 2, k \neq -2$ है।

(B) निकाय का केवल एक हल है, यदि $k = -2$ है।

(C) निकाय का केवल एक हल है, यदि $k = 2$ है।

(D) निकाय का कोई हल नहीं है, यदि $k = 2$ है।

(E) निकाय के अनन्त हल हैं, यदि $k \neq -2$ है।

तो निम्न कथनों में कौन से सत्य हैं ?

(A) केवल (C) तथा (D)

(B) केवल (A) तथा (E)

(C) केवल (B) तथा (E)

(D) केवल (A) तथा (D)

Ans. D

Sol.
$$D = \begin{vmatrix} 1 & -2 & 0 \\ 1 & -1 & k \\ 0 & k & 4 \end{vmatrix} = 4 - k^2 = (2 - k)(2 + k)$$

$$D_x = \begin{vmatrix} 1 & -2 & 0 \\ -2 & -1 & k \\ 6 & k & 4 \end{vmatrix} = -(k + 10)(k + 2)$$

$$D \neq 0 \Rightarrow k \neq \pm 2$$

$\Rightarrow$ system will have unique solution

$\Rightarrow$ A is true; E, B, C are false

$$\text{If } k = 2 \Rightarrow D = 0, D_x \neq 0$$

$\Rightarrow$ System will have no solution $\Rightarrow$ D is true

$\Rightarrow$ A & D are true.

109. The following system of linear equations :

$$2x + 3y + 2z = 9$$

$$3x + 2y + 2z = 9$$

$$x - y + 4z = 8$$

(A) Has infinitely many solutions

(B) Has a solution (α, β, γ) satisfying $\alpha + \beta^2 + \gamma^2 = 12$

(C) Does not have any solution

(D) Has a unique solution

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$$2x + 3y + 2z = 9$$

$$3x + 2y + 2z = 9$$

$$x - y + 4z = 8$$

(A) के अनन्त हल हैं

(B) का एक हल (α, β, γ) है जो $\alpha + \beta^2 + \gamma^2 = 12$ को संतुष्ट करता है

(C) का कोई हल नहीं है

(D) का केवल एक हल है

Ans. D

Sol. $\Delta = \begin{vmatrix} 2 & 3 & 2 \\ 3 & 2 & 2 \\ 1 & -1 & 4 \end{vmatrix}$

$$= 2(8 + 2) - 3(12 - 2) + 2(-3 - 2)$$

$$= 32 - 30 - 10 = -8$$

$$\Delta \neq 0$$

has a unique solution

110. Consider the following system of equations :

$$x + 2y - 3z = a$$

$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c,$$

where a, b and c are real constants. Then the system of equations :

(A) has no solution for all a, b and c

(B) has infinite number of solutions when $5a = 2b + c$

(C) has a unique solution when $5a = 2b + c$

(D) has a unique solution for all a, b and c

निम्न समीकरण निकाय पर विचार कीजिए

$$x + 2y - 3z = a$$

$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c,$$

जहाँ a, b तथा c वास्तविक अचर हैं। तो इस समीकरण निकाय :

(A) का सभी a, b तथा c के लिए कोई हल नहीं है (B) के अनन्त हल हैं जब $5a = 2b + c$

(C) का केवल एक हल है जब $5a = 2b + c$ (D) का सभी a, b तथा c के लिए केवल एक हल है

Ans. B

Sol. $D = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 6 & -11 \\ 1 & -2 & 7 \end{vmatrix}$

$$= 1(42 - 22) - 2(14 + 11) - 3(-4 - 6)$$

$$= 20 - 50 + 30 = 0$$

$$D_x = \begin{vmatrix} a & 2 & -3 \\ b & 6 & -11 \\ c & -2 & 7 \end{vmatrix}$$

$$= a[42 - 22] - b[14 - 6] + c(-22 + 18)$$

$$= 20a - 8b - 4c$$

$$= 4[5a - 2b - c]$$

$$D_y = \begin{vmatrix} 1 & a & -3 \\ 2 & b & -11 \\ 1 & c & 7 \end{vmatrix}$$

$$= -5[5a - 2b - c]$$

$$D_z = \begin{vmatrix} 1 & 2 & a \\ 2 & 6 & b \\ 1 & -2 & c \end{vmatrix}$$

$$= -2 [5a - 2b - c]$$

$$\text{for } 5a - 2b - c = 0, D_x = D_y = D_z = 0$$

∴ system has infinite number of solutions

$$\text{when } 5a = 2b + c$$

111. Let α, β, γ be the real roots of the equation $x^3 + ax^2 + bx + c = 0$, ($a, b, c \in \mathbb{R}$ and $a, b \neq 0$). If the system of equations (in u, v, w) given by $\alpha u + \beta v + \gamma w = 0$; $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is :

माना α, β, γ समीकरण $x^3 + ax^2 + bx + c = 0$, ($a, b, c \in \mathbb{R}$ तथा $a, b \neq 0$) के वास्तविक मूल हैं। यदि u, v, w में समीकरण निकाय $\alpha u + \beta v + \gamma w = 0$; $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ का अतुच्छ हल है, तो $\frac{a^2}{b}$ का मान है :

- (A) 5 (B) 3 (C) 0 (D) 1

Ans. B

Sol. For the system of equations to have a non-trivial solution.

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0$$

$$\Rightarrow \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

$$\Rightarrow \alpha = \beta = \gamma \text{ or } \alpha + \beta + \gamma = 0$$

But since $\alpha + \beta + \gamma = -a$ and $a \neq 0$

$$\Rightarrow \alpha + \beta + \gamma \neq 0$$

$$\text{So, } \alpha = \beta = \gamma$$

$$\text{So } 3\alpha = -a; \quad 3\alpha^2 = b$$

$$\frac{a^2}{b} = \frac{9\alpha^2}{3\alpha^2} = 3$$

112. Let S_1 and S_2 be respectively the sets of all $a \in \mathbb{R} - \{0\}$ for which the system of linear equations :

$$ax + 2ay - 3az = 1$$

$$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$$

$$(3a + 5)x + (a + 5)y + (a + 2)z = 3$$

has unique solution and infinitely many solutions. Then :

(A) S_1 is an infinite set and $n(S_2) = 2$ (B) $S_1 = \mathbb{R} - \{0\}$ and $S_2 = \phi$

(C) $n(S_1) = 2$ and S_2 is an infinite set (D) $S_1 = \phi$ and $S_2 = \mathbb{R} - \{0\}$

माना सभी $a \in \mathbb{R} - \{0\}$, जिनके लिए रैखिक समीकरण निकाय

$$ax + 2ay - 3az = 1$$

$$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$$

$$(3a + 5)x + (a + 5)y + (a + 2)z = 3$$

का केवल एक हल है तथा अनंत हल है, के समुच्चय क्रमशः S_1 तथा S_2 है। तो

(A) S_1 एक अपरिमित समुच्चय है तथा $n(S_2) = 2$ है

(B) $S_1 = \mathbb{R} - \{0\}$ तथा $S_2 = \phi$

(C) $n(S_1) = 2$ तथा S_2 एक अपरिमित समुच्चय है

(D) $S_1 = \phi$ तथा $S_2 = \mathbb{R} - \{0\}$

Ans. B

Sol.
$$D = \begin{bmatrix} a & 2a & -3a \\ 2a+1 & 2a+3 & 2a+1 \\ 3a+5 & a+5 & a+2 \end{bmatrix} = a(15a^2 + 31a + 37)$$

$$\Rightarrow D \neq 0 \text{ for all } a \in \mathbb{R} - \{0\}$$

113. If $1, \log_{10}(4^x - 2)$ and $\log_{10}\left(4^x + \frac{18}{5}\right)$ are in arithmetic progression for a real number x , then the value of the

determinant
$$\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$$
 is equal to :

यदि एक वास्तविक संख्या x के लिए $1, \log_{10}(4^x - 2)$ तथा $\log_{10}\left(4^x + \frac{18}{5}\right)$ एक समान्तर श्रेणी में है, तो सारणिक

$$\begin{vmatrix} 2\left(x - \frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix} \text{ का मान बराबर } \underline{\hspace{2cm}} \text{ ।}$$

Ans. 2

Sol. $2 \log_{10} (4^x - 2) = 1 + \log_{10} \left(4^x + \frac{18}{5} \right)$

$$\log_{10} (4^x - 2)^2 = \log_{10} \left(10 \left(4^x + \frac{18}{5} \right) \right)$$

$$(4x - 2)^2 = 10 \cdot 4^x + 36 \quad 4^x = y$$

$$(y - 2)^2 = 10y + 36$$

$$y^2 - 4y + 4 = 10y + 36$$

$$y^2 - 14y - 32 = 0$$

$$(y - 16)(y + 2) = 0$$

$$y = 16$$

$$y = -2, \text{ which is not possible}$$

$$4^x = 16$$

$$x = 2$$

Now, value of determinant

$$\begin{vmatrix} 3 & 1 & 4 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix}$$

$$= 3(-2) - 1(0 - 4) + 4(1)$$

$$= -6 + 4 + 4$$

$$= 2$$

Matrices

114. Let $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2 + 1 & b \\ 1 & b & 2 \end{bmatrix}$ where $b > 0$. Then the minimum value of $\frac{\det(A)}{b}$ is :
- (A) $\sqrt{3}$ (B) $-2\sqrt{3}$ (C) $-\sqrt{3}$ (D) $2\sqrt{3}$

माना $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2 + 1 & b \\ 1 & b & 2 \end{bmatrix}$ जहाँ $b > 0$ है। तो $\frac{\det(A)}{b}$ का न्यूनतम मान है

(A) $\sqrt{3}$ (B) $-2\sqrt{3}$ (C) $-\sqrt{3}$ (D) $2\sqrt{3}$

Ans. D

Sol. $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2 + 1 & b \\ 1 & b & 2 \end{bmatrix}$

$$\begin{aligned} \det(A) &= 2[2b^2 + 2 - b^2] - b(2b - b) + 1[b^2 - b^2 - 1] \\ &= 2(b^2 + 2) - b(b) + 1(-1) \\ &= 2b^2 + 4 - b^2 - 1 \\ \det(A) &= b^2 + 3 \end{aligned}$$

Let $y = \frac{\det(A)}{b} = b + \frac{3}{b}$

Apply $AM \geq GM$

$$\frac{b + \frac{3}{b}}{2} \geq \left(b \times \frac{3}{b}\right)^{1/2}$$

$$b + \frac{3}{b} \geq 2(3)^{1/2}$$

115. If $d \in \mathbb{R}$, and $A = \begin{bmatrix} -2 & 4 + d & (\sin \theta) - 2 \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2 \sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix}$, $\theta \in [0, 2\pi]$. If the minimum value of $\det(A)$ is 8, then

a value of d is :

- (A) -7 (B) -5 (C) $2(\sqrt{2} + 2)$ (D) $2(\sqrt{2} + 1)$

माना $d \in \mathbb{R}$ तथा $A = \begin{bmatrix} -2 & 4 + d & (\sin \theta) - 2 \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2 \sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix}$, $\theta \in [0, 2\pi]$ यदि $\det(A)$ का न्यूनतम मान 8 है, तो d का एक मान

है:

(A) -7

(B) -5

(C) $2(\sqrt{2} + 2)$

(D) $2(\sqrt{2} + 1)$

Ans. B

Sol. $A = \begin{vmatrix} -2 & 4+d & \sin \theta - 2 \\ 1 & \sin \theta + 2 & d \\ 5 & 2\sin \theta - d & -\sin \theta + 2 + 2d \end{vmatrix}$

$R_1 \rightarrow R_1 + R_3 - 2R_2$

$A(\theta) = \begin{vmatrix} 1 & 0 & 0 \\ 1 & \sin \theta + 2 & d \\ 5 & 2\sin \theta - d & -\sin \theta + 2 + 2d \end{vmatrix}$

$A(\theta) = (\sin \theta + 2)(2 + 2d - \sin \theta) - \sin \theta + 2 + 2d$

$A(\theta) = (d + 2)^2 - \sin^2 \theta$

$A_{\min} = (d + 2)^2 - 1 = 8$

$d + 2 = \pm 3$

$d = 1, -5$

116. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to :

(A) 10

(B) 9

(C) 15

(D) 135

यदि $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ तथा $Q = [q_{ij}]$ दो ऐसे 3×3 आव्यूह हैं, कि $Q - P^5 = I_3$ है, तो $\frac{q_{21} + q_{31}}{q_{32}}$ बराबर है -

(A) 10

(B) 9

(C) 15

(D) 135

Ans. A

Sol. $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$

$P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 3+3 & 1 & 0 \\ 9+3 \times 3+1 & 3+3 & 1 \end{bmatrix}$

$$P^5 = \begin{bmatrix} 1 & 0 & 0 \\ 15 & 1 & 0 \\ 135 & 15 & 1 \end{bmatrix}$$

$$Q - P^5 = I_3$$

$$Q = I_3 + P^5$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 15 & 2 & 0 \\ 135 & 15 & 2 \end{bmatrix}$$

$$\frac{q_{21} + q_{31}}{q_{32}} \Rightarrow 10$$

117. If $A = \begin{bmatrix} e^t & e^{-t} \cos t & e^{-t} \sin t \\ e^t & -e^{-t} \cos t - e^{-t} \sin t & -e^{-t} \sin t + e^{-t} \cos t \\ e^t & 2e^{-t} \sin t & -2e^{-t} \cos t \end{bmatrix}$ then A is :

(A) invertible only if $t = \pi$

(B) not invertible for any $t \in \mathbb{R}$

(C) invertible for all $t \in \mathbb{R}$

(D) invertible only if $t = \frac{\pi}{2}$

यदि $A = \begin{bmatrix} e^t & e^{-t} \cos t & e^{-t} \sin t \\ e^t & -e^{-t} \cos t - e^{-t} \sin t & -e^{-t} \sin t + e^{-t} \cos t \\ e^t & 2e^{-t} \sin t & -2e^{-t} \cos t \end{bmatrix}$ है तो A :

(A) व्युत्क्रमणीय है, केवल $t = \pi$ है।

(B) किसी भी $t \in \mathbb{R}$ के लिए व्युत्क्रमणीय नहीं है।

(C) सभी $t \in \mathbb{R}$ के लिए व्युत्क्रमणीय है।

(D) व्युत्क्रमणीय है, केवल तब, जब $t = \frac{\pi}{2}$ है।

Ans. C

Sol. $|A| = e^{-t} \begin{vmatrix} 1 & \cos t & \sin t \\ 1 & -\cos t - \sin t & -\sin t + \cos t \\ 1 & 2 \sin t & -2 \cos t \end{vmatrix}$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1 = \begin{vmatrix} 1 & \cos t & \sin t \\ 0 & -\cos t - \sin t & -2 \sin t + \cos t \\ 0 & 2 \sin t - \cos t & -2 \cos t - \sin t \end{vmatrix}$$

$$= (2 \cos t + \sin t)^2 + (2 \sin t - \cos t)^2$$

$$|A| = 5$$

So A is invertible for all $t \in \mathbb{R}$

118. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, (\alpha \in \mathbb{R})$ such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ then a value of α is

- (A) $\frac{\pi}{32}$ (B) $\frac{\pi}{64}$ (C) 0 (D) $\frac{\pi}{16}$

माना $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, (\alpha \in \mathbb{R})$ इस प्रकार है कि $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, तो α का एक मान है :

- (A) $\frac{\pi}{32}$ (B) $\frac{\pi}{64}$ (C) 0 (D) $\frac{\pi}{16}$

Ans. B

Sol. $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$

$$A^2 = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} = \begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$\text{Then } A^4 = A^2 \cdot A^2 = \begin{bmatrix} \cos 4\alpha & -\sin 4\alpha \\ \sin 4\alpha & \cos 4\alpha \end{bmatrix}$$

$$\text{Similarly } A^8 = A^4 \cdot A^4 = \begin{bmatrix} \cos 8\alpha & -\sin 8\alpha \\ \sin 8\alpha & \cos 8\alpha \end{bmatrix}$$

$$\text{and so on } A^{32} = \begin{bmatrix} \cos 32\alpha & -\sin 32\alpha \\ \sin 32\alpha & \cos 32\alpha \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

So $\sin 32\alpha = 1$ and $\cos 32\alpha = 0$

$$\Rightarrow 32\alpha = 2n\pi + \frac{\pi}{2} \Rightarrow \alpha = \frac{n\pi}{16} + \frac{\pi}{64} \text{ where } n \in \mathbb{Z}$$

$$\text{put } n = 0, \alpha = \frac{\pi}{64}$$

119. Let the numbers 2, b, c be in an A.P. and $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$. If $\det(A) \in [2, 16]$, then c lies in the interval :

- (A) [2, 3] (B) $(2 + 2^{3/4}, 4)$ (C) $[3, 2 + 2^{3/4}]$ (D) [4, 6]

माना संख्याएं 2, b, c एक समान्तर श्रेणी में हैं तथा $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$. यदि $\det(A) \in [2, 16]$, तो c निम्न में से किस अन्तराल में

है :

- (A) [2, 3] (B) $(2 + 2^{3/4}, 4)$ (C) $[3, 2 + 2^{3/4}]$ (D) [4, 6]

Ans. D

Sol. $|A| = \begin{vmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{vmatrix} \quad c_2 \rightarrow c_2 - c_1, c_3 \rightarrow c_3 - c_1$

$$|A| = \begin{vmatrix} 1 & 0 & 0 \\ 2 & b-2 & c-2 \\ 4 & (b-2)(b+2) & (c-2)(c+2) \end{vmatrix} = (b-2)(c-2)(c-b)$$

2, b, c are in A.P. $\Rightarrow (b-2) = (c-b) = d, c-2 = 2d$

$\Rightarrow |A| = d \cdot 2d \cdot d = 2d^3$

$\therefore |A| \in [2, 16] \Rightarrow 1 \leq d^3 \leq 8 \Rightarrow 1 \leq d \leq 2$

$4 \leq 2d + 2 \leq 6 \Rightarrow 4 \leq c \leq 6$

120. If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdot \dots \cdot \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$, then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is :

(A) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$

यदि $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdot \dots \cdot \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$ है, तो $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ का व्युत्क्रम (inverse) है :

(A) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$

Ans. B

Sol. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \dots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 1+2+3+\dots+(n-1) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \frac{(n-1)n}{2} = 78$

$\Rightarrow n = 13$

Now, inverse of $\begin{bmatrix} 1 & 13 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$